CDFD Runtime Paper 01: Axioms, State Variables, and Claim Boundaries

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May 2026

Abstract

Executive synthesis. This paper defines the public axiomatic surface of Constraint-Driven Flux Dynamics (CDFD) as implemented in `cdfd_runtime`. The framework represents a system through active flow Φ, adaptive constraint C, surface responsiveness S, retained structural memory M_s, and the operating ratio $\Psi_s = (\Phi/C)SM_s$. These variables are not presented here as empirical proof of a universal physical ontology. They are a computational grammar: a state language that can be stress-tested, falsified, and reused across the runtime, CDFL models, domain adapters, validation tools, and local discovery experiments.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Purpose and Scope

The CDFD Runtime begins from a pragmatic observation: many systems fail when drive exceeds the capacity of the medium that must carry it. A thermal channel, a metabolic pathway, a city road network, a financial clearing system, and a prebiotic mineral pore are materially different. Yet each can be represented, at a coarse level, as a relation between throughput and constraint.

This paper states the shared variables. Later papers derive dynamics, describe the runtime, explain CDFL, and connect the theory to local experiments in `cdfd_runtime/experiments`. The claim discipline is deliberately strict: runtime experiments are candidate model diagnostics and falsification targets unless independent empirical validation is provided.

3 Core State Variables

Definition 1 (Flow field Φ)**.** *The flow field $\Phi(\boldsymbol{x}, t)$ is the active drive or transported quantity of a system. Depending on context, Φ may represent heat flux, substrate flux, capital velocity, information flow, pressure drive, or field intensity.*

Definition 2 (Constraint field C)**.** *The constraint field $C(\boldsymbol{x}, t)$ is the local resistance, bottleneck, capacity limit, or structural load that opposes or shapes Φ. In CDFD, C is not fixed by default. It may adapt, relax, diffuse, stiffen, or fail.*

Definition 3 (Surface responsiveness S)**.** *The surface field $S(\boldsymbol{x}, t)$ describes how readily a boundary, interface, or medium changes its routing behavior under load. A high S can improve routing when regulated, but can amplify instability when recovery is weak.*

Definition 4 (Structural memory M_s)**.** *The memory field $M_s(\boldsymbol{x}, t)$ records retained stiffening after overload. In the engine this is the hysteresis channel: a system can remain altered after the original pulse, injury, or shock has passed.*

Definition 5 (Operating ratio Ψ_s)**.** *The adaptive operating ratio is*

$$\Psi_s(\boldsymbol{x}, t) = \frac{\Phi(\boldsymbol{x}, t)}{C(\boldsymbol{x}, t)} S(\boldsymbol{x}, t) M_s(\boldsymbol{x}, t). \tag{1}$$

Low Ψ_s indicates under-driving or over-constraint. Near-unity Ψ_s marks an actively regulated operating band. High Ψ_s indicates overload, with interpretation dependent on the modeled domain.

4 Axioms

Postulate 1 (Flow requires a medium)**.** *Every modeled flow is carried by a medium with finite effective constraint. The limiting behavior $C \to 0$ is therefore a boundary condition, not an ordinary engineering state.*

Postulate 2 (Constraint responds to stress)**.** *Sustained or rapidly changing Φ can alter C, S, and M_s. This is the reason the runtime stores fields rather than isolated scalar labels.*

Postulate 3 (Regime transitions are measurable)**.** *The useful test of CDFD is not whether CDFD language can be fitted to a system after the fact. The useful test is whether changes in Φ, C, S, and M_s predict measurable transitions before they occur.*

5 Runtime Connection

These variables are implemented in `engine/state.py`. The physics update lives primarily in `engine/physics.py`; shared finite checks, JSON-safe summaries, the scalar operating ratio, the Life Number, and bounded diagnostic helpers live in `runtime/diagnostics.py`. The command surface is exposed through `cdfd.py`, while CDFL models provide a readable way to define systems and rules.

6 Experiment and Discovery Hooks

The experiments directory provides candidate diagnostics tied to these axioms:

- `experiments/notebooks/discovery_vacuum_hysteresis.py` tests whether a high-flux pulse leaves a localized M_s residual.
- `experiments/notebooks/discovery_adaptive_surface.py` stress-tests when S amplifies rather than stabilizes overload.
- `experiments/notebooks/discovery_ool_phase.py` explores the Life Number boundary $\Lambda \approx 1$ in a prebiotic toy parameter space.
- `experiments/reports/FINAL_RELEASE_EXPERIMENT_SELECTION.md` records which outputs are suitable for public discussion and which must remain only simulation logs.

7 Claim Status

The axioms are a computational foundation and a falsifiable modeling proposal. They are not by themselves a claim that every physical, biological, or social system has been empirically validated under CDFD. A domain mapping fails if it does not produce reproducible regime thresholds, if its Ψ_s proxy adds no predictive information beyond ordinary variables, or if independent data reject its proposed flow-constraint mapping.

8 Conclusion

Paper 01 establishes the vocabulary: Φ, C, S, M_s, and Ψ_s. The rest of the runtime series asks whether this vocabulary can become useful science: executable, falsifiable, reproducible, and disciplined about the difference between a model diagnostic and an empirical discovery.

References

[1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant.* CDFD Part I Paper I, preprint, 2026.

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[4] Steve Bico Mujjabi, MD. *The Vacuum Equation of State: Deriving Torus Knot Self-Consistency from the Public CDFD Balance Law $\Psi = \Phi/C$.* CDFD Part I Paper X, preprint, 2026.

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[6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics.* Public release archive, 2026, `https://doi.org/10.5281/zenodo.20250821`.